THE E_1 SPECTRAL SEQUENCE FOR THE TOTAL SIMPLICIAL SITE WITH E_1^{P,Q} = H^Q(C_P, K_P)

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: functoriality in K (statement line 2175) rides on the functoriality of the cited Homology first-quadrant spectral sequence lemma applied to the functorial double-complex construction; the author does not expand it further. The identification Gamma(D, -) = Hom(O_D, -) used for the rows is recorded at source lines 2184-2185.

Editorial note. The label lemma-simplicial-module-cohomology-site names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The simplicial-site situation with a sheaf of rings. Same chapter, Situation \label{situation-simplicial-site}. Statement lines 2162-2164: O is a sheaf of rings on C_total and K is an object of D^+(O).
- The Cech complex for modules on the total site. Same chapter, \label{lemma-cech-complex-modules}. Proof lines 2180-2189: the horizontal arrows of the double complex come from this lemma; the rows are exact in positive degrees and evaluate to Gamma(C_total, I^q) in degree 0, using Gamma = Hom(O_D, -) on modules.
- Exactness of restriction and acyclicity of restricted injectives. Same chapter, \label{lemma-restriction-to-components-site} and \label{lemma-restriction-injective-to-component-limp}. Proof lines 2192-2196: restriction to C_p is exact, so I_p^bullet represents K_p in D(C_p); the I_p^q are totally acyclic on C_p.
- The homological-algebra machinery. Homology, Lemma \label{homology-lemma-double-complex-gives-resolution}, Derived Categories, Lemma \label{derived-lemma-leray-acyclicity}, Cohomology on Sites, Lemma \label{sites-cohomology-lemma-limp-acyclic}, Homology, Lemma \label{homology-lemma-first-quadrant-ss}. Proof lines 2190-2205: the total complex computes RGamma(C_total, K); column cohomology is computed by Leray's acyclicity; the first-quadrant spectral sequence lemma concludes.

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Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Prerequisite pointer invoked at proof line(s) 2200; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0133: sources/stacks-chow/tags.txt, source lines 1421–1421. Prerequisite pointer invoked at proof line(s) 2191; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0132: sources/stacks-chow/tags.txt, source lines 1420–1420. Prerequisite pointer invoked at proof line(s) 2204; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D9C: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2135–2135. Prerequisite pointer invoked at proof line(s) 2180; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D73: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1518–1518. Prerequisite pointer invoked at proof line(s) 2197; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 2193; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0731: sources/stacks-chow/tags.txt, source lines 9045–9045. Prerequisite pointer invoked at proof line(s) 2202; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 2162; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let $\mathcal{O}$ be a sheaf of rings. For K in $D^+(\mathcal{O})$ there is a spectral sequence $(E_r, d_r)_{r\geq 0}$ with*

$$E_1^{p,q} = H^q(\mathcal{C}_p, K_p), \quad d_1^{p,q} : E_1^{p,q} \to E_1^{p+1,q}$$

converging to $H^{p+q}(\mathcal{C}_{total}, K)$. This spectral sequence is functorial in K.

Proof. Let $\mathcal{I}^\bullet$ be a bounded below complex of injective $\mathcal{O}$-modules representing K. Consider the double complex with terms

$$A^{p,q} = \Gamma(\mathcal{C}_p, \mathcal{I}_p^q)$$

where the horizontal arrows come from Lemma 0D9C and the vertical arrows from the differentials of the complex $\mathcal{I}^\bullet$. Observe that $\Gamma(\mathcal{D}, -) = \mathrm{Hom}_{\mathcal{O}_\mathcal{D}}(\mathcal{O}_\mathcal{D}, -)$ on $Mod(\mathcal{O}_\mathcal{D})$. Hence the lemma says rows of the double complex are exact in positive degrees and evaluate to $\Gamma(\mathcal{C}_{total}, \mathcal{I}^q)$ in degree 0. Thus the total complex associated to the double complex computes $R\Gamma(\mathcal{C}_{total}, K)$ by Homology, Lemma 0133. On the other hand, since restriction to $\mathcal{C}_p$ is exact (Lemma 09WG) the complex $\mathcal{I}_p^\bullet$ represents K_p in $D(\mathcal{C}_p)$. The sheaves $\mathcal{I}_p^q$ are totally acyclic on $\mathcal{C}_p$ (Lemma 0D73). Hence the cohomology of the columns computes the groups $H^q(\mathcal{C}_p, K_p)$ by Leray's acyclicity lemma (Derived Categories, Lemma 015E) and Cohomology on Sites, Lemma 0731. We conclude by applying Homology, Lemma 0132. □